

Can Data Beat the Market? Predicting NBA Games and Evaluating Kalshi Odds

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Repository: <https://github.com/wwihono/kalshi-odds>

Summary of Questions and Results

1. **How accurately do Kalshi's pregame NBA prices represent the actual probability of each team winning?** Answer: Across 1,070 games, the prices were broadly calibrated (ECE = .028; Brier = .201), although several probability ranges showed local gaps.
2. **Can an NBA prediction model outperform Kalshi's market probabilities?** Answer: Not consistently. Validation selected logistic regression; it had higher test accuracy than Kalshi (74.6% vs. 71.8%) but worse Brier score and log loss.
3. **Under what conditions are Kalshi's NBA predictions most and least accurate?** Answer: The middle-volume quartiles were strongest and the highest-volume quartile was weakest; quoted probability also rose strongly with recent-form advantage.
4. **Would differences between the model and Kalshi's prices produce positive simulated returns?** Answer: Yes in this historical test. All three edge thresholds had positive net ROI after the stated fee assumption, but the result does not establish future profitability.

Motivation

Prediction-market prices compress many traders' information into probabilities that can be tested against observed outcomes. Those prices may still be affected by liquidity, attention, uncertainty, and unequal access to information. NBA games are a useful setting because teams play frequently and pregame factors such as recent performance, rest, and rating differences can be calculated consistently.

This project evaluates rather than promotes trading. It asks whether Kalshi's prices behave like calibrated probabilities, whether models built only from information available before tipoff add predictive value, and whether apparent model-market disagreements remain after chronological validation and estimated transaction fees. Probability quality matters independently of winner-picking accuracy because a 51% forecast and a 90% forecast make the same class prediction but express very different confidence.

Dataset

The analysis combines Basketball Reference's completed 2025-26 NBA schedule and results with Kalshi KXNBAGAME market records and one-minute candlesticks. Each analytical row represents one completed game matched to the Kalshi contract for the home team. The pregame probability is the midpoint of the last Yes bid and ask whose candle ended at least 15 minutes before scheduled tipoff (KalshiEx LLC, 2026a; Sports Reference LLC, 2026).

Collection produced 1,322 completed schedule rows and 2,720 settled team contracts. The program matched 1,294 games to a unique home-team contract; 1,070 of those had a qualifying pregame quote. One full-dollar bid/ask spread was excluded from modeling, leaving 1,069 games. The EDA table contains 60 fields, including source identifiers, prices, outcomes, timing audits, and pregame features.

Table 1

Data Coverage After Los Angeles Team Normalization Repair

Pipeline stage	Rows	Schedule share	Interpretation
Completed schedule	1,322	100.0%	Basketball Reference results
Matched games	1,294	97.9%	Unique home-team contract
EDA sample	1,070	80.9%	Qualifying pregame quote
Model sample	1,069	80.9%	After one spread exclusion

Note. The 2,720 season-market records contain both team contracts, so that count is not directly comparable with game rows.

The archived EDA omitted both Los Angeles teams because labels such as LA and Los Angeles were treated as a single identity. The repaired normalization first uses explicit labels and then reads the contract ticker when the displayed name is ambiguous. It now distinguishes the Lakers from the Clippers, represents all 30 teams, and increases the usable EDA sample from 915 to 1,070 games. Twelve rows (1.1%) lack some rolling or rest values because the involved team had no prior game in the collected season; those values are imputed only after the chronological split.

Method

Data Collection and Cleaning

The collection program follows pagination cursors for current and archived Kalshi markets, downloads candles for each market, and caches the responses. Schedule tables are parsed month by month, timestamps are converted to UTC, and games are matched by date and normalized home and away teams. Automated audits check unique game and ticker keys, score-derived winners, settlement agreement, probability bounds, bid not exceeding ask, and quote timing. This pipeline prevents current outcomes or post-tipoff quotes from entering the predictors.

EDA Measures

Calibration is summarized with accuracy, Brier score, binary log loss, and 10-bin expected calibration error (ECE). Accuracy uses a .50 decision threshold. The Brier score is the mean squared difference between probability and outcome, while log loss applies a larger penalty to confident errors; lower values are better for both. ECE weights each bin's absolute difference between mean probability and observed home-win frequency. Descriptive condition analyses compare probability bins, volume quartiles, and recent-form differences.

Pregame Features and Models

Seven home-minus-away predictors use only previously completed games: five- and 10-game win percentage, five- and 10-game mean point differential, rest days, back-to-back status, and Elo rating. Kalshi prices and current-game outcomes are excluded. Missing training-period values are median-imputed and all predictors are standardized using parameters fitted without the test period.

The planned model families are logistic regression, random forest, and gradient boosting, implemented with scikit-learn (Pedregosa et al., 2011). Logistic regression compares five L2 strengths. Random forest compares 50 or 100 estimators, depths of 3 or 5, and minimum leaf sizes of 8 or 10. Gradient boosting compares 50 or 100 stages, depths of 1 or 2, and learning rates of .05 or .10. A fixed random seed makes the ensembles reproducible.

Table 2*Best Validation Candidate in Each Model Family*

Family	Selected settings	Accuracy	Brier	Log loss
Logistic regression	L2 = 0	70.4%	.192	.569
Random forest	100 trees; depth 5; leaf 8	66.0%	.200	.586
Gradient boosting	100 stages; depth 1; rate .05	66.5%	.197	.580

Note. Validation Brier score selected logistic regression before the test period was scored.

Chronological Evaluation and Trading

Games are date-disjoint: the first 650 games (October 21-February 11) form training, the next 206 (February 12-March 22) form validation, and the last 213 (March 23-June 13) form test. After tuning, each family's best candidate is refit on the 856 development games and scored once on the same test games as Kalshi. The selected family supplies the trade probabilities.

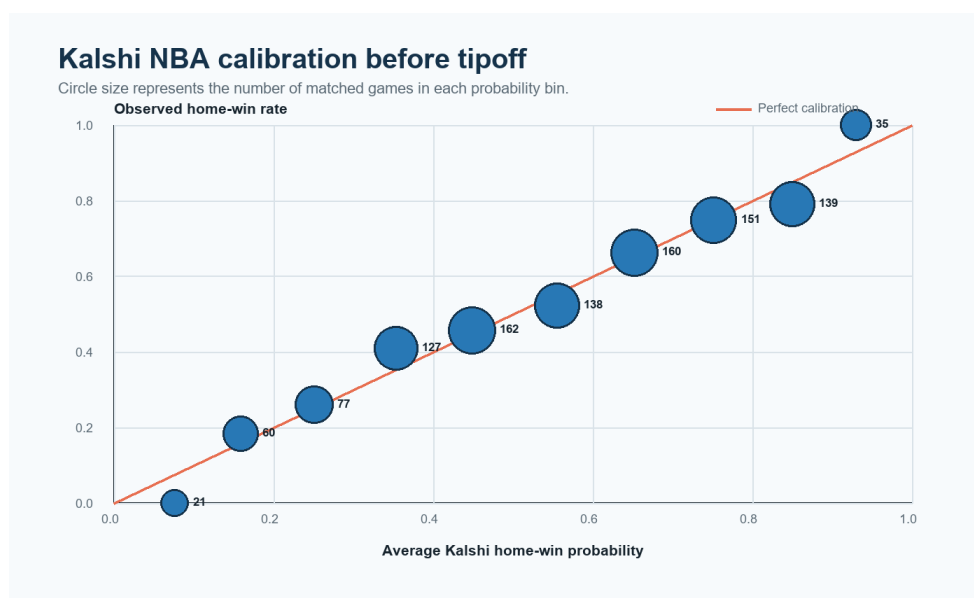
The simulation buys one home-Yes or home-No contract at the displayed ask when the selected model's advantage reaches 5%, 10%, or 15%. Profit equals settlement payout minus price and estimated taker fee. The fee is rounded up to cents from .07 times contracts times price times one minus price (KalshiEx LLC, 2026b). No real trades were placed.

Exploratory Data Analysis

EDA served two purposes: it tested whether the joined rows were credible before modeling and established the market baseline. The normalized 1,070-game sample has a mean quoted home-win probability of 55.422%, nearly identical to the 55.421% observed home-win rate. Median pregame volume was 270,858 contracts and the median spread was \$.01. Every selected quote met the pre-tipoff rule, and all Kalshi settlements agreed with the schedule-derived winner.

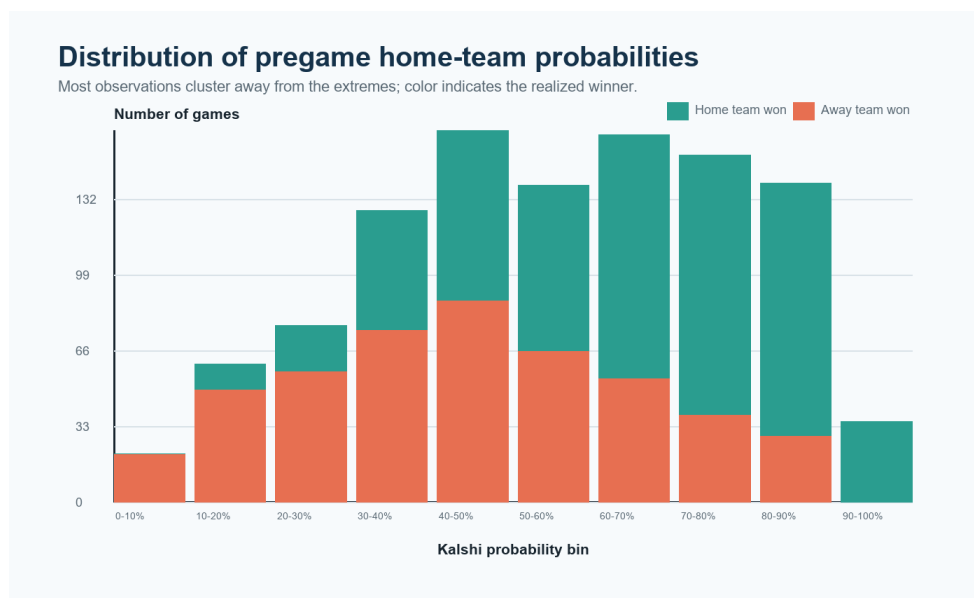
Figure 1

Calibration of Kalshi Pregame Home-Win Probabilities



Note. Generated programmatically with pandas, NumPy, and Pillow from 1,070 normalized games. The diagonal represents perfect calibration.

Figure 1 shows that most well-populated bins are close to the ideal line. The 30%-40% bin won 5.6 percentage points more often than its mean quote, while the 80%-90% bin won 5.8 points less often. The overall ECE of .028 indicates modest average bin-level error, not perfect calibration.

Figure 2*Distribution of Kalshi Pregame Home-Win Probabilities*

Note. The histogram shows the number of normalized games contributing to each probability range.

Figure 2 supplies the sample-size context for Figure 1. Only 21 games had probabilities at or below 10%, and only 35 exceeded 90%. Their large apparent calibration gaps therefore depend on far fewer outcomes than the central bins, which contain 127-162 games each.

Most importantly, EDA exposed the Los Angeles normalization failure by comparing represented teams, unmatched rows, and ticker components. Repairing that issue before modeling changed all downstream sample sizes, summaries, figures, chronological periods, test metrics, and simulated positions. That experience reinforced the value of category-coverage audits even when aggregate results initially appear plausible.

Results

Research Question 1: Kalshi Calibration

Across the full EDA sample, Kalshi achieved 67.9% directional accuracy, a .201 Brier score, and .586 log loss. The mean quote and observed home-win rate differed by less than .01 percentage point. Combined with ECE = .028, these measures support the conclusion that prices were broadly calibrated overall. Figure 1 also shows why that statement must remain aggregate: the market underpriced the 30%-40% range and overestimated the 80%-90% range in this sample.

Research Question 2: Model Performance

Logistic regression had the lowest validation Brier score and was selected without using test outcomes. Table 3 reports all three refitted families on the held-out 213-game test period. Logistic regression was the strongest model on all three metrics. Gradient boosting ranked second on Brier score and log loss, while random forest ranked second on accuracy.

Table 3

Held-Out Test Performance on 213 Games

Predictor	Accuracy	Brier score	Log loss
Logistic regression	74.6%	.178	.534
Random forest	73.2%	.189	.564
Gradient boosting	71.8%	.185	.552
Kalshi midpoint	71.8%	.175	.516

Note. Higher accuracy is better; lower Brier score and log loss are better. Hyperparameters were selected before this test period was evaluated.

The selected logistic model classified about six additional test games correctly relative to Kalshi, a 2.8-percentage-point accuracy difference. Kalshi nevertheless had a .003 lower Brier score and .018 lower log loss. This is possible because accuracy ignores confidence once a forecast falls on the correct side of .50, whereas proper scoring rules evaluate the full probability. The evidence therefore does not support a general claim that the model beat the market. It supports the narrower claim that logistic regression made better binary classifications while Kalshi supplied better probability estimates.

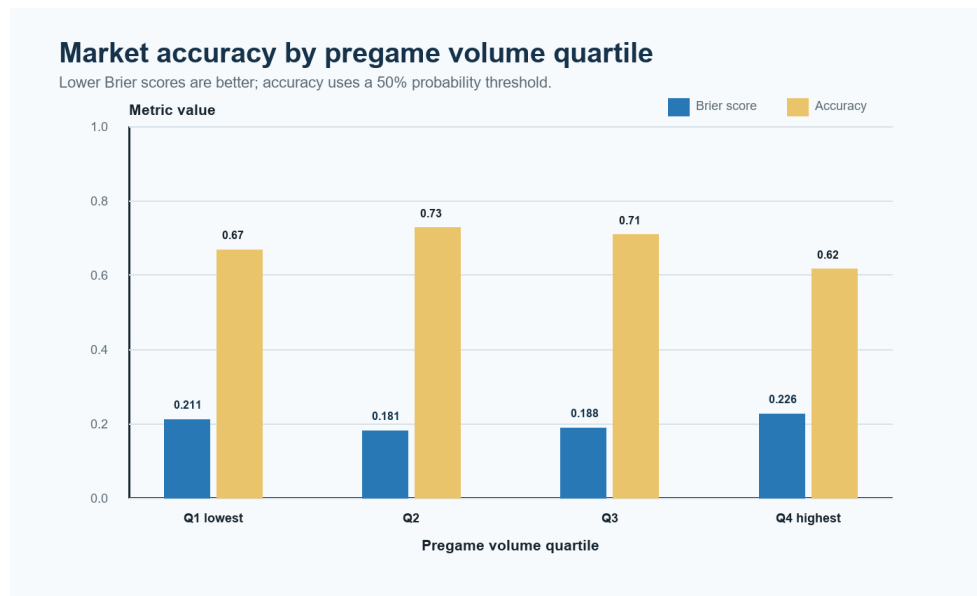
The logistic coefficients provide a limited descriptive check after standardization. Elo difference (.566) and 10-game point-differential difference (.492) had the largest positive coefficients. The five- and 10-game versions of similar measures are correlated, so individual coefficient signs should not be interpreted causally or as independent feature importance.

Research Question 3: Conditions Affecting Accuracy

Volume was not monotonically associated with more accurate prices. The second quartile had the lowest Brier score (.181) and highest accuracy (72.7%), followed by the third quartile. The highest-volume quartile had the worst Brier score (.226) and lowest accuracy (61.6%). Figure 3 displays both measures. High volume may reflect attention to unusually uncertain or news-sensitive games, so this pattern is an association rather than evidence that trading volume causes error.

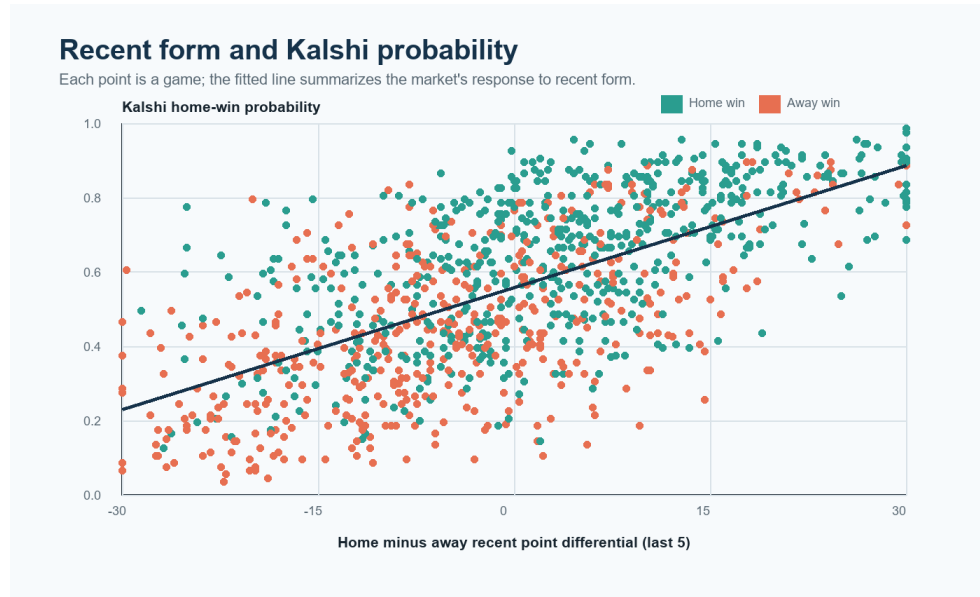
Figure 3

Accuracy and Brier Score by Pregame Volume



Note. Games are split into four nearly equal volume groups. Higher accuracy and lower Brier score indicate better performance.

Recent team form was strongly associated with price. Across 1,058 complete rows, the correlation between five-game point-differential advantage and Kalshi probability was .652. The fitted line in Figure 4 increases about 1.09 probability points for each additional point of recent differential advantage. The remaining vertical spread shows that prices also incorporate home advantage, longer-run strength, news, and other information not represented by this single feature.

Figure 4*Recent Form Difference and Kalshi Probability*

Note. Each point is one game with complete prior-five-game data; the line is an ordinary least-squares descriptive fit.

Research Question 4: Simulated Returns

All three edge thresholds produced positive historical net returns from the selected logistic model on the test period under the immediate-fill and one-contract assumptions. Larger thresholds selected fewer positions but had higher point-estimate ROI. Table 4 shows that the 15% rule returned 33.4% on 38 positions, while the 5% rule returned 11.3% on 135 positions.

Table 4*Held-Out Threshold Trade Simulation*

Minimum edge	Positions	Wins	Net profit	ROI
5%	135	54	\$5.49	11.3%
10%	73	32	\$5.08	18.9%
15%	38	17	\$4.26	33.4%

Note. Profit is net of the stated estimated taker fee. The lowest cumulative results were -\$4.94, -\$2.90, and -\$1.01 for the 5%, 10%, and 15% rules, respectively.

The position win rates are below 50% because many trades purchase lower-priced contracts whose wins pay more than their losses cost. Net profit relative to total price plus fees is therefore more informative than win rate alone. Even so, the result is not evidence of a

durable strategy. The 15% estimate depends on only 38 positions, and every threshold experienced an interim drawdown. Execution at a displayed ask, model uncertainty, selection effects, and repeated use of the same historical period can all make realized performance worse.

Impact and Limitations

This analysis may benefit students, sports analysts, and prediction-market researchers who want a reproducible example of calibration, chronological validation, and transaction-cost-aware simulation. It demonstrates why probability metrics should accompany accuracy and why category-level data audits can matter as much as model complexity.

The evidence covers one NBA season and only games with matchable contracts and qualifying quotes. Those exclusions may not be random. Scheduled and actual tipoff can differ, pregame volume covers the available quote window rather than every historical trade, and Kalshi may incorporate injuries or news absent from the feature table. The model comparison uses a limited parameter grid and one chronological split, so it does not establish performance across seasons or changing market conditions.

The simulation assumes immediate fills at displayed asks and omits queue position, latency, slippage, rejected orders, limits, changing fee schedules, capital constraints, taxes, and market impact. Potential harms include overconfidence, financial loss, and reinforcement of gambling behavior. Access is also unequal across jurisdiction, age, income, technology, and information. The conclusions should be used to study forecast evaluation, not to recommend real-money trades.

Challenge Goals

The messy-data goal was completed through paginated current and historical APIs, nested JSON, per-market candlesticks, monthly HTML schedule tables, UTC conversion, ticker-aware team normalization, audited joins, and time-restricted quote selection. The Los Angeles repair demonstrates the challenge: a plausible aggregate dataset silently omitted two teams until coverage tests connected ambiguous display labels to ticker abbreviations.

The machine-learning goal was also completed as proposed. The final pipeline compares logistic regression, random forest, and gradient boosting, tests multiple parameter combinations within each family, selects the winner using validation Brier score, refits each best candidate on the development period, and evaluates all families once on a later test period. The

implementation uses scikit-learn, a library covered in CSE 163, rather than a custom model library.

Work Plan Evaluation

The proposal allocated 24 total hours. Exact actual hours were not logged, so the evaluation below compares relative effort rather than inventing precise totals. Collection and feature engineering were near the anticipated sequence, while cleaning and verification exceeded expectations because the first team-normalization result looked valid until all-team coverage was checked. Model work expanded back to the proposed three families after the normalized data were stable.

Table 5

Proposed Work Plan Compared With Implementation

Task	Proposed	Evaluation
Collect and inspect	3 h	Longer: split APIs and per-market candles
Clean and merge	5 h	Longer: LA ambiguity required ticker logic
Pregame features	5 h	Close: rolling, rest, and Elo completed
Model evaluation	6 h	Expanded: three tuned families plus holdout
Visuals and report	5 h	Four figures and full rubric report completed

Note. Evaluations are qualitative because no reliable time log was maintained.

The estimate closest to reality was feature construction because its rolling, rest, and Elo requirements were known in advance. Cleaning was least predictable because source labels created a silent category-coverage error rather than a program crash. A better future plan would reserve a separate audit block before EDA and require expected-category, unmatched-row, and before-versus-after coverage checks before downstream analysis begins.

Testing

Two runnable assertion-based programs contain 26 test functions: five analysis tests and 21 model tests. Small deterministic DataFrames make expected outcomes inspectable by hand. Analysis tests cover team normalization, market filtering, chronological feature construction, unique processed keys, quote timing, settlement agreement, and exact Brier-score

reproduction. The LA tests include explicit Lakers and Clippers labels plus ambiguous display names resolved by ticker.

Model tests cover cleaning audits, date-disjoint splitting, median imputation, standardization, all three family trainers and predictors, metric formulas, regularization and family selection, fitted-model summaries, taker fees, position logic, threshold summaries, the complete in-memory workflow, and the file-writing main function. Every model-module function has a same-named test. Fixed random seeds make ensemble results reproducible.

End-to-end checks require valid probabilities, test-only model predictions, no ambiguous LA labels, qualifying pregame times, settlement agreement, and model-ready row counts matching the cleaning audit. On August 16, 2026, both test programs passed, generated-output checks passed, and flake8 reported no violations. These tests support trust in program behavior and reproduced calculations, but they cannot prove that the external source data are complete or that historical results will generalize.

Citations and Collaboration

The entire project was completed individually. The core analysis libraries were covered in class. Pillow was used to generate figures, and lxml supported HTML schedule parsing. I used a variety of outside sources, such as public repositories, API documentation, library documentation, and open forums, to consult and work through my project.

AI disclosure. OpenAI Codex was used to fix stylistic issues and to check, run, and validate testing outputs. I also used it to help brainstorm data normalization for the Los Angeles teams problem mentioned above. Codex also suggested using ThreadPoolExecutor to speed up I/O-bound API collection while limiting the number of active worker threads.

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